

AI-Powered Multi-Modal Content Generation with Adaptive Voice Cloning and Lip Synchronization

Project Team

Sarim Farooq 22i-0458
Amin Fahim 22i-0618
Musab Waseem 22i-0488

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Supervised by

Mr. Muhammad Aamir Gulzar

Co-Supervised by

Dr. Muhammad Asif Naeem



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

Introduction

This proposal outlines an AI-powered system designed to streamline personalized digital content creation. The system generates realistic, high-quality video outputs from simple audio or short video inputs, with native bilingual support for Urdu and English. The core pipeline integrates speech-to-text transcription, language model-based text refinement, adaptive voice cloning that improves over time, and lip synchronization technology to produce polished, single-speaker videos [7]. By consolidating these capabilities into a unified workflow, the system makes professional content creation accessible to non-technical users, reducing the complexity and cost traditionally associated with video production [6].

1.1 Problem Statement

Content creation today is time-consuming, skill-intensive, and often requires expensive tooling and software. While various AI-powered tools exist for isolated steps such as transcription, text-to-speech conversion, or video editing, there is currently no single integrated solution that addresses the complete workflow for bilingual Urdu and English content [1]. Existing tools lack native Urdu support and force users to switch between multiple platforms, leading to inefficiencies and quality inconsistencies. There is a critical gap in solutions that can natively handle bilingual inputs, refine scripts intelligently using advanced language models, generate adaptive cloned voices that improve with continued use [4], and produce lip-synced videos against user-provided templates. This fragmented ecosystem requires significant technical expertise and manual intervention at each stage. Our proposed system addresses these challenges by providing a coherent, end-to-end pipeline optimized for single-speaker content generation, enabling creators, educators, influencers, and businesses to produce professional-quality videos without specialized skills or extensive resources.

1.2 Scope

The initial release of this system focuses on single-speaker, template-based video generation with comprehensive support for both Urdu and English languages. Users will be able to upload audio clips or short video files of up to five minutes in duration. The system encompasses several key functionalities: performing accurate speech-to-text transcription for both Urdu and English audio inputs, refining and structuring the transcribed text using advanced language models to improve clarity and coherence, implementing adaptive voice cloning technology that learns from user inputs and progressively improves voice quality over time, synchronizing generated speech with facial movements in user-provided video templates through lip-sync technology, and exporting polished, ready-to-publish video outputs in standard formats. The system will maintain a user profile management system to store voice samples, preferred templates, and historical data for continuous improvement. It is explicitly designed to minimize technical overhead while maintaining high output quality. Features that are out-of-scope for the initial version include multi-speaker dialogues, advanced avatar animation beyond template-based lip synchronization, and large-scale distributed training infrastructure.

1.3 Modules

The proposed system is decomposed into seven interconnected modules, each with clearly defined responsibilities to ensure modularity and maintainability.

1.3.1 Module 1: User and Profile Management

This module provides a minimal but secure account management system to persist user settings, voice samples, and preferred video templates. It handles user registration and authentication with industry-standard security practices, manages profile-linked datasets including audio samples, text transcripts, and video templates, and implements consent tracking mechanisms for voice data usage and storage to ensure privacy compliance.

1.3.2 Module 2: Speech-to-Text and Text Refinement

This module handles bilingual transcription and intelligent script enhancement. It performs accurate Urdu and English speech-to-text conversion using state-of-the-art models [5], applies language model-based text cleanup to improve formatting and coherence, generates structured summaries and enhanced versions of transcripts, and handles mixed-

language inputs without relying on Roman transliteration where possible to maintain linguistic authenticity.

1.3.3 Module 3: Adaptive Voice Cloning

This module implements progressive voice synthesis technology that improves with user interaction. It experiments with few-shot, zero-shot, and hybrid voice cloning approaches to determine optimal performance, establishes voice quality evaluation metrics including Mean Opinion Score (MOS) assessments, implements a user approval feedback loop for continuous improvement, and maintains persistent per-user voice models that evolve with additional training data.

1.3.4 Module 4: Lip Synchronization

This module synchronizes generated speech with facial movements in user-provided video templates. It integrates and compares multiple lip-sync models including Wav2Lip [3] and alternative approaches, implements alignment quality checks and adjustment heuristics to ensure natural-looking results, and provides optional enhancement and upscaling capabilities using post-processing techniques such as RealESRGAN [8] when needed for quality improvement.

1.3.5 Module 5: Video Generation and Template Management

This module handles template-based video assembly and export functionality. It manages a library of video templates with customizable backgrounds and overlay options, implements a robust audio-video composition and rendering pipeline, and exports final outputs to standard formats suitable for direct publishing on various platforms.

1.3.6 Module 6: User Interface and User Experience

This module provides accessible and intuitive front-end workflows for all user interactions. It implements streamlined upload, review, and export interfaces, provides comprehensive error handling and user guidance throughout the workflow, and supports iterative user experience improvements based on feedback and usability testing.

1.3.7 Module 7: Deployment and Evaluation

This module focuses on system operationalization and quality assurance. It establishes packaging, continuous integration/continuous deployment (CI/CD), and deployment pipelines, implements latency monitoring and audio quality checks using MOS-style metrics, measures intelligibility and naturalness of outputs, and maintains comprehensive test plans, documentation, and demonstration materials.

1.4 User Classes and Characteristics

The system is designed to serve multiple distinct user classes, each with specific needs and characteristics.

Table 1.1: User Classes and their Characteristics	
User Class	Description
Content Creators	Digital content creators who produce regular video content for platforms such as YouTube, TikTok, or Instagram. These users require fast turnaround times, consistent quality, and bilingual support for Urdu and English audiences. They typically have moderate technical skills and need an intuitive interface that minimizes production time while maintaining professional output quality.
Educators	Teachers, trainers, and educational content developers who create instructional videos and learning materials. These users value clarity of speech, accurate transcription for subtitles, and the ability to generate personalized educational content efficiently. They may have limited technical expertise and require straightforward workflows with minimal learning curve.
Influencers	Social media influencers and brand ambassadors who need to produce engaging, personalized video content at scale. These users prioritize authentic voice representation, quick iteration cycles, and the ability to maintain consistent personal branding across multiple video outputs. They expect mobile-friendly workflows and rapid content generation.
Business Users	Marketing professionals, corporate communicators, and small business owners who require professional video content for advertising, training, or customer communication. These users need reliable, high-quality outputs with template customization options and the ability to generate bilingual content for diverse markets.
Project Development Team	The internal team of developers, testers, and designers responsible for implementing, testing, and maintaining the system. This team requires comprehensive documentation, modular architecture, and clear API specifications for efficient development and debugging.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the proposed AI-powered content generation system.

2.1 Use-case Diagram

The system follows a use case-driven approach given its interactive, end-user-focused nature. The use case diagram illustrates the primary interactions between different user classes and the system functionalities.

The primary actors include Content Creators, Educators, Influencers, and Business Users who interact with the system through registration, media upload, voice cloning, text refinement, lip-sync video generation, and content export functionalities. The system administrator manages user accounts, templates, and system configuration.

2.2 Functional Requirements

This section describes the functional requirements organized by system modules, expressed in natural language for clarity and traceability.

2.2.1 Module 1: User and Profile Management

Following are the functional requirements for user and profile management:

1. The system shall allow new users to register with email and password authentication.

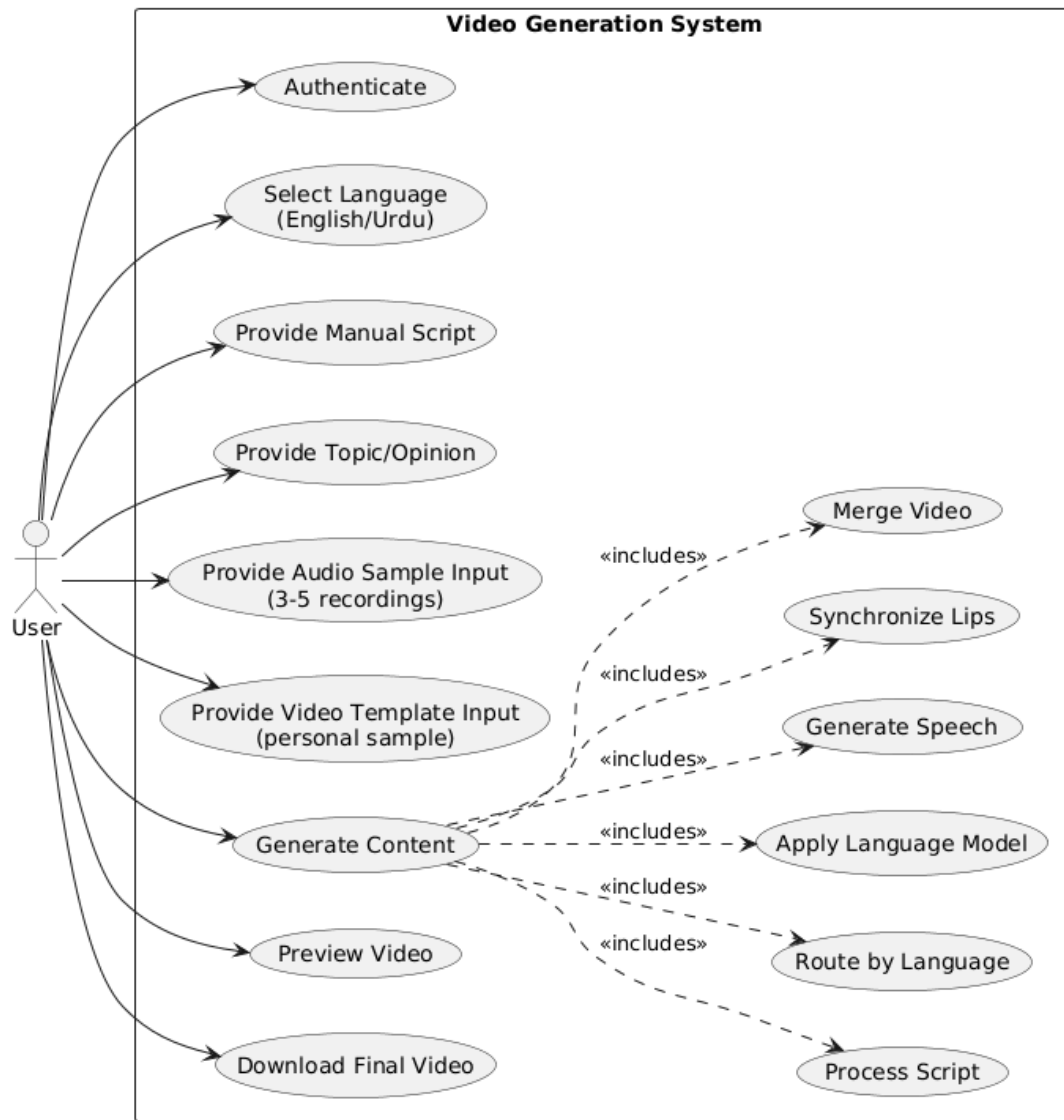


Figure 2.1: Use Case Diagram showing primary user interactions

2. The system shall securely store user credentials using industry-standard encryption.
3. The system shall maintain user profiles containing voice samples, preferred templates, and historical data.
4. The system shall implement session management with automatic logout after inactivity.
5. The system shall track and store user consent for voice data usage and storage.
6. The system shall allow users to view, update, and delete their profile information.

2.2.2 Module 2: Speech-to-Text and Text Refinement

Following are the functional requirements for STT and text processing:

1. The system shall transcribe Urdu and English audio with minimum 85% accuracy.
2. The system shall support mixed-language audio inputs without requiring transliteration.
3. The system shall automatically detect the language of input audio.
4. The system shall refine transcribed text using language models to improve structure and clarity.
5. The system shall generate formatted scripts with proper punctuation and paragraph breaks.
6. The system shall allow users to review and edit refined transcripts before processing.
7. The system shall store both original and refined transcripts for user reference.

2.2.3 Module 3: Adaptive Voice Cloning

Following are the functional requirements for voice cloning functionality:

1. The system shall generate synthetic speech that closely matches the user's voice characteristics.
2. The system shall support few-shot voice cloning with minimal input samples (3-5 recordings).
3. The system shall implement zero-shot voice cloning for rapid prototyping.
4. The system shall maintain persistent user voice models that improve with additional samples.
5. The system shall allow users to approve or reject generated voice outputs.
6. The system shall calculate and display voice quality metrics for user feedback.
7. The system shall support both Urdu and English speech synthesis in cloned voices.

2.2.4 Module 4: Lip Synchronization

Following are the functional requirements for lip-sync functionality:

1. The system shall synchronize generated audio with facial movements in user-provided videos.
2. The system shall support multiple video template formats (MP4, AVI, MOV).
3. The system shall detect and track facial landmarks for accurate lip synchronization.
4. The system shall implement quality checks to ensure natural-looking lip movements.
5. The system shall provide optional video enhancement and upscaling.
6. The system shall handle videos with varying resolutions and aspect ratios.

2.2.5 Module 5: Video Generation and Template Management

Following are the functional requirements for video generation:

1. The system shall maintain a library of customizable video templates.
2. The system shall allow users to upload custom video templates.
3. The system shall compose audio and video streams into final output files.
4. The system shall export videos in standard formats (MP4, MOV) suitable for publishing.
5. The system shall support background replacement and overlay customization.
6. The system shall render videos with configurable quality settings (720p, 1080p).

2.2.6 Module 6: User Interface

Following are the functional requirements for the user interface:

1. The system shall provide a web-based interface accessible via modern browsers.
2. The system shall implement drag-and-drop functionality for media uploads.
3. The system shall display real-time progress indicators during processing.

4. The system shall provide preview functionality for generated content before final export.
5. The system shall implement comprehensive error messages with suggested solutions.
6. The system shall support responsive design for desktop and tablet devices.

2.2.7 Module 7: Deployment and Evaluation

Following are the functional requirements for deployment and testing:

1. The system shall implement automated testing for all core functionalities.
2. The system shall monitor and log system performance metrics.
3. The system shall provide API documentation for all backend services.
4. The system shall implement continuous integration and deployment pipelines.
5. The system shall generate quality assessment reports for audio and video outputs.

2.3 Non-Functional Requirements

This section specifies quality attributes and constraints that the system must satisfy.

2.3.1 Reliability

REL-1: The system shall maintain 95% uptime during operational hours, with mean time between failures (MTBF) exceeding 720 hours under normal load conditions.

REL-2: The system shall implement automatic recovery mechanisms for transient failures in transcription and voice cloning modules.

REL-3: The system shall provide graceful degradation when third-party services are unavailable, notifying users of reduced functionality.

REL-4: The system shall persist user data and processing state to prevent data loss during unexpected shutdowns.

2.3.2 Usability

USE-1: The system shall allow users to initiate content generation with no more than three distinct interactions from the main dashboard.

USE-2: The system shall provide context-sensitive help and tooltips for all major features.

USE-3: New users shall be able to generate their first video output within 15 minutes of account creation, including voice sample collection.

USE-4: The system shall support undo/redo functionality for text editing operations.

USE-5: Error messages shall be displayed in both English and Urdu based on user preference.

2.3.3 Performance

PER-1: The system shall transcribe up to 5 minutes of audio within 30 seconds on standard hardware.

PER-2: Text refinement using language models shall complete within 10 seconds for typical input lengths (500-1000 words).

PER-3: Voice cloning inference shall generate speech at a rate of at least 1:1 (real-time) relative to audio duration.

PER-4: Lip synchronization processing shall complete within 2 minutes for 1 minute of source video at 1080p resolution.

PER-5: The web interface shall load initial page content within 2 seconds over a 10 Mbps connection.

PER-6: The system shall support concurrent processing of up to 10 user requests without significant performance degradation.

2.3.4 Security

SEC-1: The system shall encrypt all user passwords using bcrypt or equivalent hashing with minimum 12-round salting.

SEC-2: The system shall implement HTTPS for all client-server communications.

SEC-3: The system shall sanitize all user inputs to prevent SQL injection and cross-site scripting attacks.

SEC-4: User voice samples and generated content shall be stored with user-specific access controls.

SEC-5: The system shall implement API rate limiting to prevent abuse (100 requests per hour per user).

SEC-6: The system shall log all authentication attempts and flag suspicious activities.

2.3.5 Scalability

SCA-1: The system architecture shall support horizontal scaling to accommodate increasing user loads.

SCA-2: Database design shall accommodate at least 10,000 registered users with minimal performance impact.

SCA-3: Media storage shall be designed to handle growth to 1TB within the first year of operation.

2.3.6 Maintainability

MAI-1: The codebase shall maintain minimum 70% unit test coverage for core modules.

MAI-2: All modules shall follow consistent coding standards and documentation practices.

MAI-3: The system shall implement structured logging for debugging and monitoring.

2.4 Domain Model

The domain model represents the key entities and their relationships within the system.

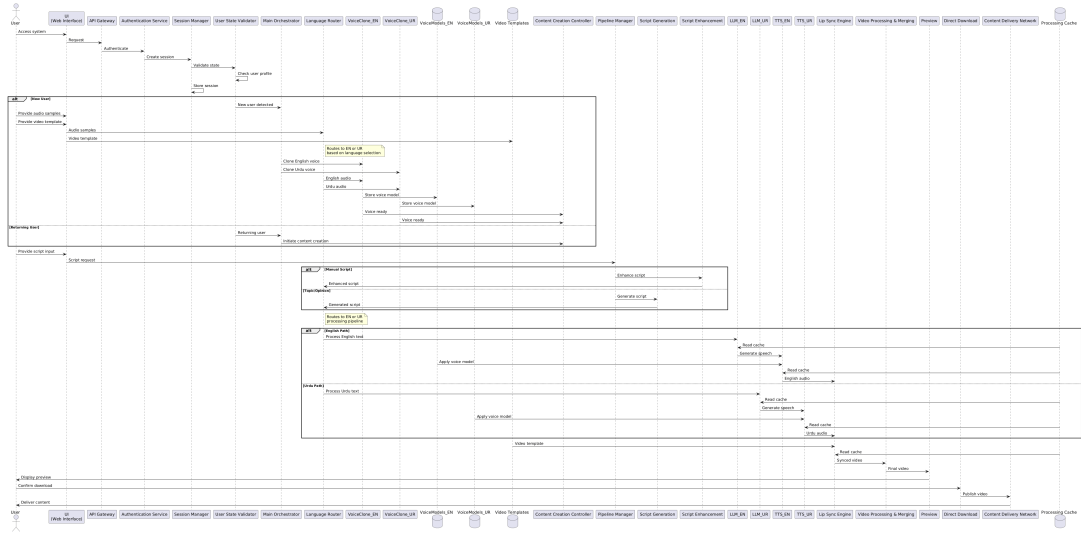


Figure 2.2: Sequence Diagram illustrating the workflow from upload to final video generation

The core entities include: User (containing profile, credentials, preferences), Audio Sample (raw input, transcription, metadata), Voice Model (cloned voice parameters, quality metrics), Text Content (original transcript, refined script, formatting rules), Video Template (template file, configuration, background assets), and Generated Video (final output, rendering settings, quality metrics). Relationships are established through user ownership, processing pipelines, and template associations.

Presentation Layer: This layer handles all user interactions through a responsive web-based interface. It implements the UI/UX module (Module 6) and provides forms for media upload, profile management, content preview, and export functionality. The presentation layer communicates with the business logic layer through RESTful API endpoints.

Business Logic Layer: This is the core processing layer containing Modules 2-5. It orchestrates the workflow from audio input to final video output. Key components include:

- **STT and Text Refinement Service:** Processes audio inputs using Whisper or equivalent models for transcription, then applies language model-based refinement
- **Voice Cloning Service:** Manages few-shot and zero-shot voice model training, inference, and quality evaluation
- **Lip Sync Service:** Implements Wav2Lip and alternative models for facial synchronization
- **Video Generation Service:** Handles template management, audio-video composition, and rendering

Data Layer: This layer manages all persistent storage including user profiles (Module 1), voice models, media files, and system metadata. It implements secure database operations and file storage with appropriate access controls.

The modules interact through well-defined interfaces:

1. User uploads audio/video through the Presentation Layer
2. User & Profile Management authenticates and validates the request
3. STT & Text Refinement processes the audio and generates refined transcript
4. Adaptive Voice Cloning generates synthetic speech using the user's cloned voice
5. Lip Synchronization aligns the generated audio with video templates
6. Video Generation composes and renders the final output
7. The completed video is returned to the user for preview and export

This modular decomposition enables parallel development, facilitates testing, and supports future extensions such as multi-speaker support or advanced avatar animation.

3.2 Design Models

This section presents detailed design models that capture the system's behavior, data flow, and component interactions.

3.2.1 Data Flow Diagram - Level 0 (Context Diagram)

The context diagram provides a high-level view of the system and its interactions with external entities.

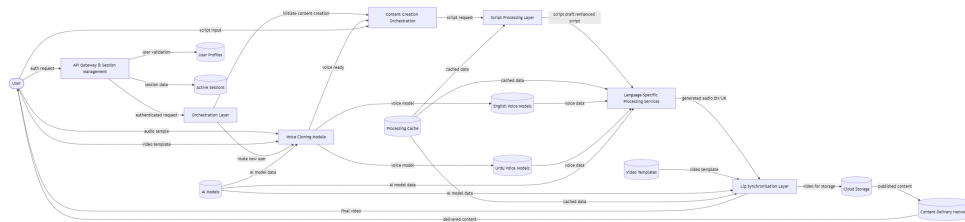


Figure 3.2: Data Flow Diagram Level 0 - Context Diagram

The Level 0 DFD shows the system as a single process that accepts audio/video inputs, user credentials, and template selections from users, and produces refined transcripts, synthetic speech, and lip-synced videos as outputs.

3.2.2 Data Flow Diagram - Level 1

The Level 1 diagram decomposes the system into major processes and data stores.

The Level 1 DFD illustrates:

- **Process 1: User Authentication & Management** - Validates users and manages profiles
- **Process 2: Speech-to-Text Processing** - Transcribes audio to text
- **Process 3: Text Refinement** - Enhances transcript quality using language models

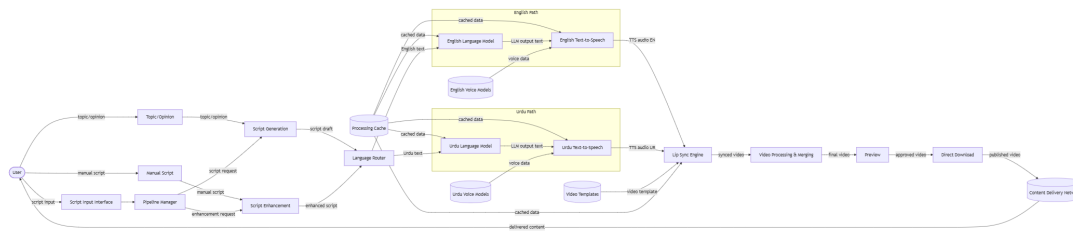


Figure 3.3: Data Flow Diagram Level 1 - Major Processes

- **Process 4: Voice Model Training & Inference** - Clones user voice and generates speech
- **Process 5: Lip Synchronization** - Aligns audio with facial movements
- **Process 6: Video Rendering & Export** - Produces final video output

Data stores include User Database, Media Storage, Voice Model Repository, Template Library, and Output Cache.

3.2.3 Deployment Diagram

The deployment diagram illustrates the physical distribution of system components across hardware nodes.

The deployment architecture consists of:

- **Client Devices:** Web browsers on user devices (desktop, tablet) accessing the application via HTTPS
- **Web Server:** Hosts the frontend application and serves static assets
- **Application Server:** Runs backend services including API endpoints, business logic, and ML model inference
- **Database Server:** Stores user data, metadata, and system configuration

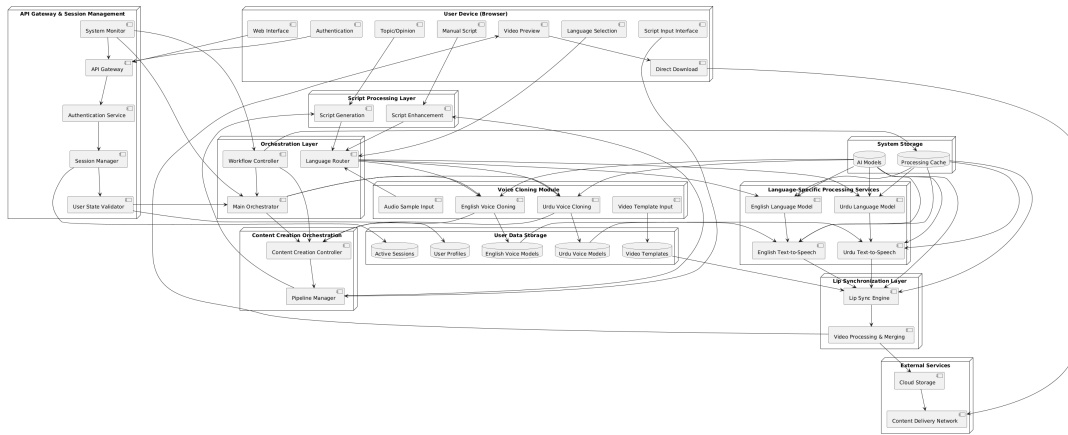


Figure 3.4: Deployment Diagram showing component distribution

- **File Storage:** Cloud or local storage for media files, voice models, and generated videos
- **ML Processing Node:** Optional GPU-accelerated server for compute-intensive tasks (voice cloning, lip-sync)

Communication between nodes occurs over secure protocols with appropriate load balancing and caching mechanisms.

3.3 Data Design

This section describes how the information domain is transformed into data structures and how data is stored, processed, and organized within the system.

3.3.1 Database Schema

The system uses a relational database to manage structured data with the following primary entities:

Users Table:

- **user_id (Primary Key):** Unique identifier
- **email:** User email address (unique)
- **password_hash:** Encrypted password
- **created_at, updated_at:** Timestamps

- language_preference: Default language (Urdu/English)
- consent_voice_storage: Boolean flag for consent

Audio_Samples Table:

- sample_id (Primary Key): Unique identifier
- user_id (Foreign Key): Reference to Users
- file_path: Storage location of audio file
- duration: Audio length in seconds
- language: Detected language
- upload_timestamp: Upload date and time
- transcription: Generated text transcript

Voice_Models Table:

- model_id (Primary Key): Unique identifier
- user_id (Foreign Key): Reference to Users
- model_path: Storage location of trained model
- training_samples: Number of samples used for training
- quality_score: Evaluated quality metric (MOS)
- version: Model version number for tracking improvements
- created_at: Training timestamp

Text_Contents Table:

- content_id (Primary Key): Unique identifier
- user_id (Foreign Key): Reference to Users
- original_transcript: Raw STT output
- refined_script: Language model enhanced text
- language: Content language

- word_count: Number of words
- created_at: Creation timestamp

Video_Templates Table:

- template_id (Primary Key): Unique identifier
- template_name: Descriptive name
- file_path: Storage location of template video
- resolution: Video resolution (e.g., 1920x1080)
- duration: Template duration in seconds
- thumbnail_path: Preview image location
- is_public: Boolean flag for shared templates

Generated_Videos Table:

- video_id (Primary Key): Unique identifier
- user_id (Foreign Key): Reference to Users
- template_id (Foreign Key): Reference to Video_Templates
- voice_model_id (Foreign Key): Reference to Voice_Models
- content_id (Foreign Key): Reference to Text_Contents
- output_path: Storage location of final video
- rendering_settings: JSON object with quality parameters
- processing_time: Time taken to generate video
- created_at: Generation timestamp

3.3.2 File Storage Organization

Media files and ML models are stored in a hierarchical file system structure:


```
/storage/  
  /users/  
    /{user_id}/  
      /audio/          # Raw audio uploads  
      /videos/         # Generated videos  
      /voice_models/   # User-specific voice models  
  /templates/  
    /public/           # Shared video templates  
    /private/          # User-uploaded templates  
  /cache/  
    /temp/             # Temporary processing files  
    /thumbnails/       # Preview images
```

Files are named using UUIDs to prevent conflicts and ensure uniqueness. Metadata is stored in the database with references to file paths.

3.3.3 Data Processing Pipeline

Data flows through the system in the following sequence:

1. **Input Stage:** Raw audio/video uploaded and stored; metadata created in database
2. **Transcription Stage:** Audio processed by STT model; transcript stored in Text_Contents
3. **Refinement Stage:** Transcript enhanced by language model; updated in database
4. **Voice Synthesis Stage:** Refined text converted to speech using cloned voice model
5. **Lip Sync Stage:** Generated audio synchronized with video template
6. **Output Stage:** Final video rendered, stored, and database record created

All processing stages implement error handling and rollback mechanisms to ensure data consistency. Processing state is tracked through status fields in database records, enabling resume functionality for interrupted operations.

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